

# How Tomasulo's Algorithm Prevents a WAW Hazard

A Write-After-Write (WAW) hazard occurs when two instructions write to the same destination register and the later instruction completes before the earlier one. Without special handling, the earlier instruction could overwrite the newer value, corrupting the architectural register file.

Consider the following two instructions from Part A:

I2: MUL.D F4, F2, F6

I4: SUB.D F4, F8, F2

Both instructions write to register F4, creating a WAW hazard.

**Cycle 2: Issue of I2 :** Instruction I2 is issued to reservation station Mult1. Since I2 will eventually write F4, the Register Status Table is updated:

| Register | Producer |
|----------|----------|
| F4       | Mult1    |

This indicates that the newest version of F4 is expected from reservation station Mult1. Any future instruction needing F4 will refer to this tag.

## Cycle 4: Issue of I4

Instruction I4 is in reservation station Add2. I4 also writes to F4. At issue time, Tomasulo performs register renaming by updating the Register Status Table:

| Register | Producer |
|----------|----------|
| F4       | Add2     |

The previous entry (Mult1) is overwritten by Add2. This is the step that prevents the WAW hazard. The processor now considers Add2 to be the producer of the most recent version of F4.

The reservation station for I4 contains:

| Busy | Op  | Vj | Vk | Qj   | Qk |
|------|-----|----|----|------|----|
| Yes  | SUB | —  | F2 | Add1 | —  |

Since F8 is not yet available, I4 waits for Add1 to produce it.

### Cycle 9: I4 Completes

After receiving all operands, I4 executes and broadcasts its result on the Common Data Bus. The result is written to F4. Because the Register Status Table still points to Add2, this write is accepted as the current architectural value of F4.

The Register Status entry for F4 is then cleared, indicating that the latest value has been committed.

### Cycle 13: I2 Completes

The multiplication instruction I2 finally completes and broadcasts its result from Mult1. However, before writing the result, Tomasulo checks the Register Status Table.

At this point, the table no longer contains:

F4 → Mult1

Instead, F4 has already been updated by I4. Therefore, the result from Mult1 is recognized as obsolete. Although the value may appear on the CDB, it is not allowed to overwrite the architectural register F4.

Thus, Tomasulo's algorithm prevents WAW hazards through register renaming. When I4 is issued, the Register Status Table entry for F4 is changed from Mult1 to Add2. As a result, when the older instruction (I2) eventually completes, its result is ignored because it is no longer the current producer of F4. This ensures that the architectural register file always contains the value from the most recently issued writer, preventing corruption caused by out-of-order completion.